

# Three-chromatic (0,2)-graphs

Unrefereed candidate proof by GPT-6 Astra

Writeup: gpt-6-astra (single model pass)

Catalog id `three_chromatic_0_2_graphs` — generated 2026-09-16

**UNREFEREED MODEL OUTPUT.** This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim below as unverified.

## Summary and provenance

|                        |                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Catalog id             | <code>three_chromatic_0_2_graphs</code>                                                                                                     |
| Catalog entry          | <a href="#">Three-chromatic (0,2)-graphs</a>                                                                                                |
| Source corpus          | opg                                                                                                                                         |
| Campaign leg           | OpenProblemGarden first pass ( <code>attacks_opg</code> )                                                                                   |
| Source paper / entry   | <a href="http://www.openproblemgarden.org/op/three_chromatic_0_2_graphs">http://www.openproblemgarden.org/op/three_chromatic_0_2_graphs</a> |
| Model verdict          | <b>proved</b> (confidence: high)                                                                                                            |
| Model's one-line claim | Every finite 3-colourable (0,2)-graph is bipartite, so none has chromatic number exactly three.                                             |
| Model                  | gpt-6-astra, reasoning effort max, mode=pro, flex service tier                                                                              |
| Original artifact      | <code>attacks_opg/three_chromatic_0_2_graphs/output.md</code>                                                                               |
| Independent review     | <b>None</b>                                                                                                                                 |
| Model's caveats        | For finite simple undirected graphs; literature priority has not been independently checked.                                                |

## Problem statement

Question Are there any (0,2)-graphs with chromatic number exactly three?

## Catalog context

A (0,2)-graph is a graph such that every pair of distinct vertices has either 0 or 2 common neighbours. It is fairly easy to see that a (0,2)-graph is necessarily regular and a variety of other properties can be shown to hold. Although (0,2)-graphs with chromatic number 2, 4 and 5 are known it is open as to whether there can be any (0,2)-graphs with chromatic number exactly three.

## Astra writeup

*The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.*

## Statement

**Theorem.** Let  $G$  be a finite simple graph in which every pair of distinct vertices has either zero or two common neighbours. If  $G$  admits a proper colouring with three colours, then  $G$  is bipartite.

The proof uses a neighbourhood-matching property to show that every real-valued edge function whose sum around each quadrangle is zero is a vertex-potential difference. A proper three-colouring supplies precisely such an edge function.

### 1. The neighbourhood-matching property

For an edge  $uv$ , put

$$A = N(u) \setminus \{v\}, \quad B = N(v) \setminus \{u\}.$$

For each  $a \in A$ , the vertices  $v, a$  have the common neighbour  $u$ , and hence have a unique second common neighbour  $b$ . Define  $\phi(a) = b$ . Then  $b \in B$ ,  $ab \in E(G)$ , and

$$u, v, b, a, u$$

is a quadrangle with four distinct vertices.

This map is a bijection. Indeed,  $u, b$  have  $v, a$  as their two common neighbours. Consequently, the analogous construction from  $B$  to  $A$ —taking the common neighbour of  $u, b$  other than  $v$ —is its inverse.

Thus, for every edge  $uv$ , there is a bijection

$$\phi : N(u) \setminus \{v\} \longrightarrow N(v) \setminus \{u\} \quad \text{with} \quad a\phi(a) \in E(G). \quad (1)$$

In particular, adjacent vertices have equal degrees, so each connected component is regular. The matching property itself, rather than regularity, is what we need.

### 2. A lemma about edge functions

An **antisymmetric edge function** assigns a real number  $\omega(x, y)$  to every ordered adjacent pair, with

$$\omega(y, x) = -\omega(x, y).$$

Say it is **square-closed** if its sum around every quadrangle is zero.

**Lemma.** Suppose a finite connected graph satisfies (1). Every square-closed antisymmetric edge function  $\omega$  has the form

$$\omega(x, y) = h(y) - h(x) \quad (2)$$

for some  $h : V(G) \rightarrow \mathbb{R}$ .

**Proof.** Define

$$\operatorname{div} \omega(x) = \sum_{y \in N(x)} \omega(x, y), \quad \Delta h(x) = \sum_{y \in N(x)} (h(y) - h(x)).$$

Antisymmetry gives

$$\sum_x \operatorname{div} \omega(x) = 0.$$

The Laplacian of a finite connected graph has image equal to the space of zero-sum functions. For completeness, it is symmetric, and

$$\langle h, -\Delta h \rangle = \sum_{\{x, y\} \in E(G)} (h(x) - h(y))^2$$

shows that its kernel consists exactly of the constants. Hence there exists  $h$  satisfying

$$\Delta h = \operatorname{div} \omega.$$

Set

$$\eta(x, y) = \omega(x, y) - (h(y) - h(x)).$$

Then  $\eta$  is antisymmetric, square-closed, and divergence-free:

$$\sum_{y \in N(x)} \eta(x, y) = 0 \quad \text{for every } x. \quad (3)$$

Suppose  $\eta$  is not identically zero. By finiteness and antisymmetry, its maximum over ordered edges is some  $M > 0$ . Choose an ordered edge  $uv$  with

$$\eta(u, v) = M.$$

Use the matching (1). For  $a \in A$ , write  $b = \phi(a)$ . Square-closedness on  $u, v, b, a, u$  gives

$$\eta(u, v) + \eta(v, b) - \eta(a, b) - \eta(u, a) = 0.$$

Therefore

$$\eta(v, b) - \eta(u, a) = \eta(a, b) - M \leq 0.$$

Summing over the matching yields

$$\sum_{b \in B} \eta(v, b) \leq \sum_{a \in A} \eta(u, a). \quad (4)$$

But (3) and antisymmetry identify the two sides as

$$\sum_{b \in B} \eta(v, b) = -\eta(v, u) = M, \quad \sum_{a \in A} \eta(u, a) = -\eta(u, v) = -M.$$

Thus (4) says  $M \leq -M$ , contradicting  $M > 0$ .

Consequently  $\eta = 0$ , proving (2).  $\square$

### 3. Applying the lemma to a three-colouring

Work in a connected component of  $G$ ; a one-vertex component is harmless. Let

$$c : V(G) \rightarrow \mathbb{Z}/3\mathbb{Z}$$

be a proper colouring. For each ordered edge  $xy$ , define  $\omega(x, y)$  to be the unique number in  $\{-1, 1\}$  satisfying

$$\omega(x, y) \equiv c(y) - c(x) \pmod{3}. \quad (5)$$

This is well-defined and antisymmetric.

Consider any quadrangle. The sum of  $\omega$  around it is:

- divisible by 3, by telescoping (5); and
- an even integer between  $-4$  and  $4$ , being a sum of four numbers from  $\{-1, 1\}$ .

The only possibility is zero. Thus  $\omega$  is square-closed.

The lemma now provides  $h : V(G) \rightarrow \mathbb{R}$  such that, on every edge,

$$h(y) - h(x) = \omega(x, y) \in \{-1, 1\}. \quad (6)$$

Subtract a constant so that  $h(r) = 0$  at one chosen vertex. Connectivity and (6) then imply that every  $h(x)$  is an integer. Moreover, adjacent vertices have opposite parities. Hence

$$x \mapsto h(x) \pmod{2}$$

is a proper two-colouring.

Applying this independently to every component proves that  $G$  is bipartite. Therefore **no finite  $(0, 2)$ -graph has chromatic number exactly three.**  $\square$

### Scope and gaps

All finite simple graphs in the question are covered, including disconnected graphs by the componentwise argument. No Cayley-graph or cube-like hypothesis is used, and there is no remaining mathematical case in the proof. I have not independently checked whether this argument or the resulting theorem has already appeared in the literature.